Homework 1 Report
Data Driven Business Analytics (MGT-302)

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1 Linear Regression

1.1 Part 1

Task: Exclude Mont Ventoux and Pic du Midi, fit a linear regression with latitude, longitude, altitude, and discuss coefficient relevance.

Answer. Using the reduced dataset (without Mont Ventoux and Pic du Midi), the reported linear model coefficients are:

$$\beta = [37.2650,\ -0.53396,\ 0.03210,\ -0.006414]$$

where the vector corresponds to intercept and the spatial predictors used in the model.

The reported p-values indicate that not all three spatial attributes are equally informative. In particular, longitude has a high p-value (≈ 0.4236), while latitude and altitude are more significant. Therefore, the model selection step keeps the two strongest predictors and removes the less informative one.

1.2 Part 2

Task: Build model with the two most significant attributes, predict Mont Ventoux and Pic du Midi temperatures, and include 95% confidence intervals.

Answer. The reduced model (using the two most significant variables) reports:

$$\text{MSE} = 0.4815989858138969$$

Predictions and confidence intervals:

- **Mont Ventoux**: true temperature = 3.6, predicted = 6.1876, 95% CI = $(4.3337,\ 8.0414)$.
- **Pic du Midi**: true temperature = -1.2, predicted = -3.4637, 95% CI = $(-8.1348,\ 1.2073)$.

Interpretation of uncertainty:

- For Mont Ventoux, the measured value is outside the 95% interval, suggesting the model under-represents this extreme case.
- For Pic du Midi, the measured value lies inside the 95% interval, indicating consistency with model uncertainty for this point.

1.3 Part 3

Task: Report MSE and discuss 3D scatterplot quality.

Answer. The reported mean squared error of the second model is:

$$\text{MSE} = 0.4815989858138969$$

From the provided interpretation, the model quality is good overall for regular observations, but weaker for extreme stations (especially high mountain conditions). This indicates that the linear relation captures the dominant trend, while local extremes may require richer features or a non-linear model to improve fit.

The 3D scatterplot can be generated from the project code using: `python main.py` in the `LinearRegression` folder.

2 Principal Component Analysis (PCA)

2.1 1. Data Preprocessing

Task: Present centered data.

Answer. TODO

2.2 2. Covariance Structure

Task: Present covariance/scatter matrix.

Answer. TODO

2.3 3. Eigen Decomposition

Task: Derive eigenvalues of covariance matrix.

Answer. TODO

2.4 4. Projection

Task: Project data onto first principal component.

Answer. TODO

2.5 5. Interpretation

Task: Discuss redundancy, intrinsic dimensionality, and explained variance.

Answer. TODO

3 K-means Clustering

3.1 Part 1

Task: Explain initialization sensitivity and one mitigation strategy.

Answer. K-means minimizes the within-cluster sum of squares (WCSS), but this objective is non-convex. As a result, the algorithm may converge to different local minima depending on the initial centroid positions. Two runs with different random starts can therefore produce different cluster assignments and different final WCSS values.

A robust mitigation is **K-means++ initialization**, which spreads initial centroids by selecting each new centroid with probability proportional to the squared distance from the closest already chosen centroid. This reduces poor starts and generally improves stability. Another standard practice is multiple random restarts and selecting the run with the lowest WCSS.

3.2 Part 2

Task: Implement K-means with K=3 and visualize clusters + centroids.

Answer. The dataset from `data_cluster.csv` was loaded and clustered using a custom NumPy implementation with iterative assignment and centroid update steps until convergence.

For $K = 3$, the final centroids obtained were:

- Cluster 0: $[4.9753,\ 6.1344]$
- Cluster 1: $[0.8643,\ 0.9280]$
- Cluster 2: $[8.1211,\ 1.9222]$

The cluster visualization uses color-coded points and explicit centroid markers, with labeled axes and legend for readability.

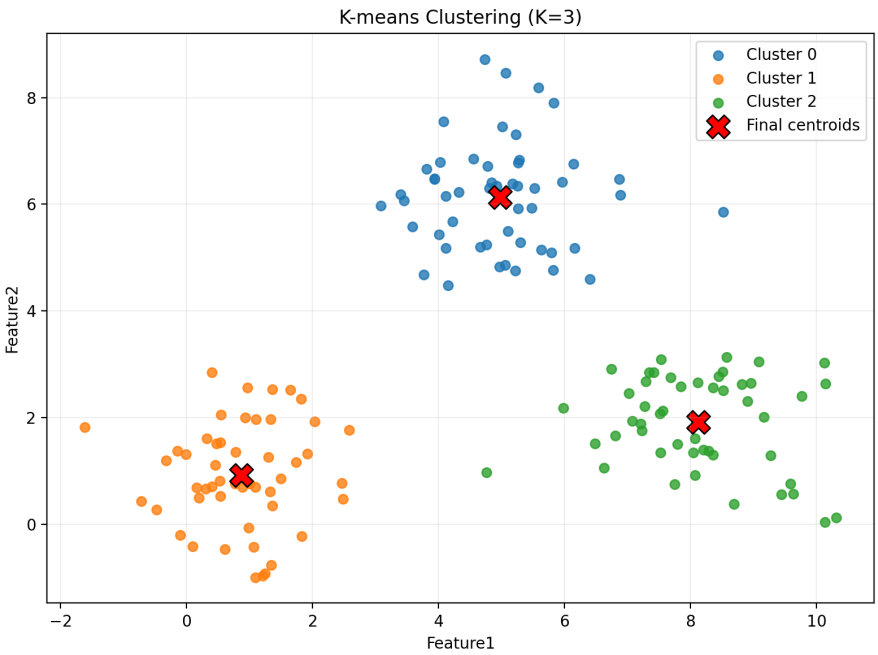

Figure 1: K-means clusters for K=3.

3.3 Part 3

Task: Build elbow curve (K=2..6) and interpret optimal K.

Answer. WCSS values for the elbow analysis are:

K	WCSS
2	974.4308
3	283.7533
4	252.0666
5	227.9890
6	185.4181

The elbow is clearly around $K = 3$: there is a very large improvement from $K = 2$ to $K = 3$, followed by progressively smaller gains for larger K. This aligns with the expected three-cluster structure of the synthetic data.

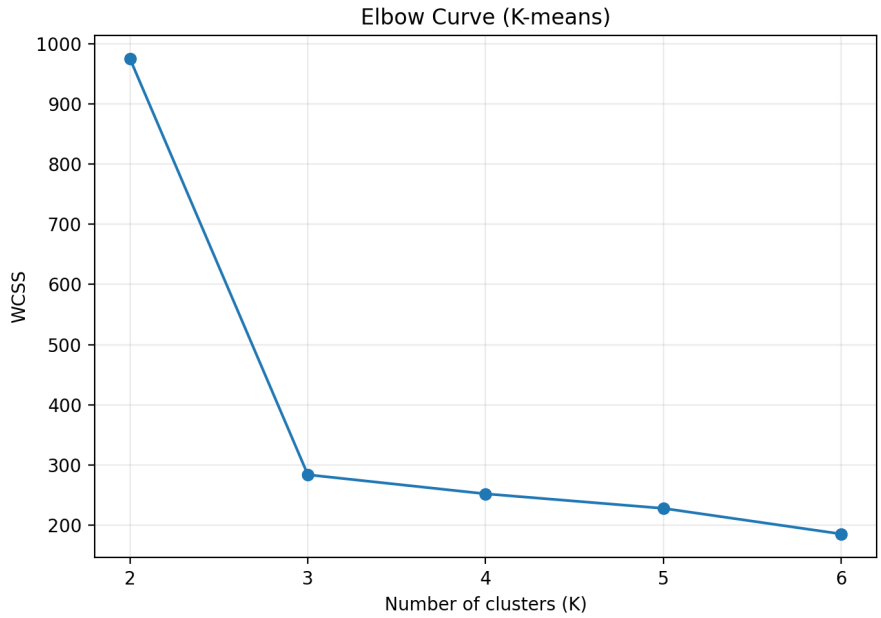

Figure 2: Elbow curve for K-means (WCSS vs K).

3.4 Part 4

Task: Explain WCSS trend with K and discuss initialization effects.

Answer. WCSS decreases monotonically as K increases because additional centroids provide more flexibility: each point can be assigned to an equal or closer center compared to smaller K. Therefore, the minimum achievable WCSS cannot increase when a new cluster is allowed.

To quantify initialization sensitivity, $K = 3$ was tested with seeds 0 to 9:

Seed	Iterations to Converge	Final WCSS
0	4	283.7533
1	4	283.7533
2	5	943.2848
3	4	946.9119
4	4	283.7533
5	3	283.7533
6	4	283.7533
7	4	283.7533
8	5	283.7533
9	4	283.7533

Most initializations converge to the best solution (WCSS ≈ 283.75), but some seeds converge to much worse local minima (WCSS $\approx$ 943 to 947). This directly confirms the theoretical sensitivity to centroid initialization.

4 Support Vector Machine (SVM)

4.1 1. Review of SVM Strategies

Task: Summarize OvR vs OvO and choose suitable strategy.

Answer. Support Vector Machines are fundamentally binary classifiers, but multi-class classification can be handled through decomposition strategies:

- **One-vs-Rest (OvR):** train K classifiers for K classes; each classifier separates one class from all remaining classes.
- **One-vs-One (OvO):** train $\frac{K(K-1)}{2}$ pairwise classifiers and aggregate predictions by voting.

In scikit-learn, `LinearSVC` typically uses OvR, while `SVC` uses OvO internally. For the digits dataset, both are valid: OvR is computationally efficient, while OvO with kernels can better model nonlinear boundaries.

4.2 2. Regularization Parameter

Task: Explain role of C and other tunable parameters.

Answer. The soft-margin parameter C controls the bias-variance trade-off:

- **Small C:** stronger regularization, wider margin, more tolerated training errors, risk of underfitting.
- **Large C:** weaker regularization, fewer tolerated errors, narrower margin, risk of overfitting.

For RBF SVM, γ is also critical:

- **High γ:** highly flexible and localized decision boundary.
- **Low γ:** smoother and more global boundary.

Good performance requires tuning both C and (for kernels) γ.

4.3 3. Linear SVM Experiment

Task: Train/validate over logarithmic C grid and discuss validation accuracy.

Answer. With an 80%/20% train-validation split on the digits dataset and C values swept logarithmically from 10^{-4} to 10^{4}, the best reported linear-SVM validation accuracy is:

$$0.9694 \quad (96.94\%) \text{ at } C = 0.1.$$

For very small C, regularization is too strong and the model underfits. As C increases, accuracy improves and reaches a maximum near $C = 0.1$. Beyond that point, performance slightly decreases and stabilizes around 0.947, consistent with mild overfitting for overly large C.

4.4 4. Kernel SVM Experiment

Task: Repeat with RBF kernel and compare to linear SVM.

Answer. Repeating the same C sweep with an RBF kernel yields low accuracy for very small C (around 20%), then strong improvements as C increases. The best reported result is:

$$0.9806 \quad (98.06\%) \text{ at } C = 10.$$

Comparison with linear SVM:

- Linear best: 0.9694
- RBF best: 0.9806

The RBF model performs better, indicating that the digits data is not perfectly linearly separable. This improvement is explained by the kernel trick, which implicitly maps samples to a higher-dimensional feature space where more complex class boundaries can be represented.